PASSIVT MÅLEKORT UDKAST – IKKE 4 kW EFFEKTTRIN
Ingen forbindelse til 230 V AC eller PE. Flydende DC–sekundaer AGND.

J1
POS_SENSE
NEG_SENSE
VOLTAGE SENSE 0..120V

R1
POS_SENSE
VP1
130k 0.1%

R2
VP1
VP2
130k 0.1%

R3
VP2
VS_PLUS
130k 0.1%

R4
VS_PLUS
10k 0.1%
AGND

R9
VS_PLUS
VMON_PLUS
1k 1%

C1
VMON_PLUS
10nF C0G 50V
AGND

R5
NEG_SENSE
VN1
130k 0.1%

R6
VN1
VN2
130k 0.1%

R7
VN2
VS_MINUS
130k 0.1%

R8
VS_MINUS
10k 0.1%
AGND

R10
VS_MINUS
VMON_MINUS
1k 1%

C2
VMON_MINUS
10nF C0G 50V
AGND

Vout = 40*(VMON_PLUS – VMON_MINUS), ideal unloaded ratio.
Both sense inputs: 0..120 V relative AGND; no negative common mode.
130 k resistors rated >=100 V; max 39 V drop at 120 V total input.

J2
KELVIN_PLUS
KELVIN_MINUS
AGND
EXTERNAL SHUNT KELVIN

R11
KELVIN_PLUS
IMON_PLUS
1k 1%

C3
IMON_PLUS
10nF C0G 50V
IMON_MINUS

R12
KELVIN_MINUS
IMON_MINUS
1k 1%

J3
AGND
VMON_PLUS
VMON_MINUS
IMON_PLUS
IMON_MINUS
AGND
MONITOR OUTPUT – NO ADC

External 1 mohm four–terminal shunt >=10 W, rated >=50 A DC; outside PCB.
40 A –> 40 mV; I = (IMON_PLUS – IMON_MINUS) / 0.001 ohm.
TP1/TP2 energized HV: no hand probing; guarded fixture only.

TP1 POS_SENSE
TP2 NEG_SENSE
TP3 AGND
TP4 VS_PLUS
TP5 VS_MINUS
TP6 VMON_PLUS
TP7 VMON_MINUS
TP8 KELVIN_PLUS
TP9 KELVIN_MINUS
TP10 IMON_PLUS
TP11 IMON_MINUS

No active protection, no ADC, no CC/CV control.
Monitor loads affect gain; input impedance >=10 Mohm.
Do not connect unpowered ADC. Prototype only.

No ADC or clamps; input >=10 Mohm; powered frontend only
External four–terminal 1 mohm shunt; 40 A gives 40 mV
0..120 V relative AGND; divider 1/40; 100 V gives 2.5 V
Floating DC secondary; NONISOLATED to secondary AGND

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Title: MÅLEKORT UDKAST – IKKE 4 kW EFFEKTTRIN

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 1/1